15.12

wall me kis jagah transverse rf kahan nahi denge?

(1) Vert rf kidhese km h?

(2) compression side me vert rf hi nai dethe?

Note: Transverse reinforcement is not provided along with vertical reinforcement if area of vertical reinforcement is not more than 0.02 times of gross area ($A_v > 0.01 A_g$)

or where the vertical reinforcement is not required for compression.

Wksht 7 - 1c 4 7c. 9c

10 d

3 April 2024 Class 16

Miscallaneous provisions

TMT : thermomechanically treated

7 days to 28 days strength

approx ratio = $\frac{2}{3}$

$\frac{\text{7 days str}}{\text{28 days str}} = \frac{2}{3} = 0.66 \simeq 0.65$ ✓

maximum possible value of fresh [green] concrete = 1.0

What is the full form of TMT?

What is 7days str ratio to 28 days str

Maximum possible value of compaction factor for green concrete

YES 2001

Q

A Slump test → 1 300-500 mm

B Compaction factor test → 2 75 to 125 mm

C Vee bee consistometer → 3 0.80-0.96

D flow test → 4 0 to 10 sec

kyon flow test me zyada mm milta h slump se?

Answer: slump test depth me result deta hai.

- compaction factor ratio ya decimal me result deta hai
- Vee bee time me result deta hai
- flow test flow width (mm/cm) me result deta hai

16.2 I Es 1998

Batching flexible way me kaam krta h ya controlled way me

kyon batching karteh?

Transporting and handling kya hotah?

kaise kaste kaig?

kya ways hai?

transport kote time kya dhyan me rakhnahai (rokna)?

Q Batching refers to controlling the quantity of each material into each batch. the measurement of material for making concrete is called batching. It is essential to ensure uniformity of proportions and aggregate grading in successive batches.

Transporting and handling after mixing concrete will be transported to formwork as rapidly as possible by methods which will prevent the segregation.

Common way of transportation is given below.

1) Head load
2) Dumper
3) Transit mixer [RMC ready mix concrete]
4) Concrete pump

Concrete ka macala kitne height segira sakte hain?

Batching handling mixing placing transportn ka sequence of operation kya hai? IES 2006

Curing kya hota hai?

Lab curing imp h

kyo types hai curing ke

Maximum permissible free fall of concrete may be taken as 1.5 metre. (if free fall is more than this then there may be possibility of segregation)

correct sequence of operation involved in concrete production

1. Batching
2. Mixing
3. Handling
4. transporting
5. Placing.
6. Compaction.

B Bhilai
M mein
H Hariraj
T Type
P place
C CCmeh

Curing: Curing is the process of preventing loss of moisture from concrete while maintaining a satisfactory temperature regime.

The prevention of moisture loss from the concrete is particularly important if water cement ratio is particularly low.

16.4] types of curing kya hai?

Moist curing kya hota hai?

kaise krte hai?

kya categories of curing hai?

Steam moist curing kiski dvara kiy caring?

Membrane curing kya hai?

Membrane curing kab kaste hai?

Membrane aur moist me fark kya hai?

Types of curing

1) Moist curing: exposed surface of concrete shall be kept continuously in damp or wet condition by ponding or by covering with layer. Steam curing is a type ~~of~~ of moist curing.

2) Membrane curing: Curing compound may be used. It is applied to all exposed surface of concrete as soon as possible the concrete has set. Example by cover.

Curtailment of tension reinforcement in flexural members.

1) for curtailment reinforcement shall extend beyond the point at which it is not longer required to resist flexure for a distance equal to effective depth of member or 12 ϕ whichever is greater except at simple supports or end of cantilever.

2) flexural reinforcement should not be terminated in tension zone unless any one of the following criteria is satisfied

a) the shear at the cutoff point does not exceed two thirds that permitted including the shear strength of web reinforcement provided.

kya hota
bhai curtai
ment of rf.
kyon krte
h?
kis member
me krte h?
kahaan par
kateng e?
fix aur continuous
beam me curtailment
ka kya rule h?
flexural
rule
tension
zone me
ka b terminate
karenge
a) shear
cutoff
wala kya pta
b) shear aur
torsion ka
kya criteria
c) 36mm wdd ka
h?

16.6

(b) in case of shear & torsion the terminated cutoff point is equal to $\frac{3}{4}^{th}$ of l_d

(c) for 36 mm & smaller bars the continuing bar provides double the area required for flexure at the cutoff point and the shear does not exceed three fourths that permitted.

Temperature and shrinkage steel is provided in RC slab because it occupies larger area.

Slab kis type ka member hai IES 2004 ka Qn.

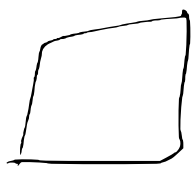
Pg 4.1 Worksheet 7 ka Qn no 2.

A S/s RC beam has clear span 5m & support width 300mm. Cross sec is 300x450mm rectangular. the effective span of beam is.

Ans 2

l_o : 5m = 5000 mm

W : 300 mm



300 x 450 mm

l_{eff} : ?

S/s beam.

Solution.

(i) S/s beam

(a) if $W \leq \frac{l_o}{12}$

$\leq \frac{5000}{12}$

300 mm

≤ 416

$l_{eff} = l_o + W$ } ↓
$l_o + d$ ↓ D

$= 5000 + 300 = 5300$ mm } ↓
$5000 + 450 = 5450$ mm

$\Rightarrow$ Ans. $= l_{eff} = 5300$ mm.

Worksheet 7
Question 6
Numerical)
A two way RCC slab is simply supported on all edges with corner free to lift. Area of steel in short span is 412 mm². Determine torsional reinforcement at corners if required.

Ans 6. $\frac{3}{4} \times 412$

if corners are free to lift then torsional reinforcement is required to be provided:

$A_{ST} = \frac{3}{4} A_{st}$ (+ve) → 412 mm²

$\Rightarrow A_{ST} = 309$ mm²

16.8

Worksheet 7

Q no 5 soln

A cantilever beam has span 4m has cross section 150x 500mm. if checked for lateral stability & deflection the beam will fail in?

soln → (5) lateral stability

a) S/S $l_{eff} \not> 60b$

$\not> \frac{250b^2}{d}$ ↓

b) cantilever $\not> 25b$

$\not> \frac{100b^2}{d}$ ↓

l_{eff} = 4m or 400 mm

$\not> 25 \times 150$ = 3750 mm

$\not> \frac{100 \times 150^2}{500}$ = 4500 mm

↓

4000mm $\not>$ 3750mm

4500mm ↓

l_{eff} calculated se zyada nahi ho skta agar zyada ho toh unsafe hai ya beam fail ho skti laterally

l_{eff} 3750 se zyada nahi ho skta agar zyada ho toh unsafe hai ya fail hai lateral stability criteria mein.

Jab naidiya h effective depth toh fota l k. effective mach lete hain.

Check for deflection:

1) Cantelivered $\left(\frac{l}{d}\right) = 7$

$$\frac{400}{500} = 8$$

l/d ki value 7 se kam ya barabar hona chahiye tabhi safe hoga but is case me ya n me 8 aya jo ki 7 se zyada hai matlab unsafe ya fails in deflection criteria.

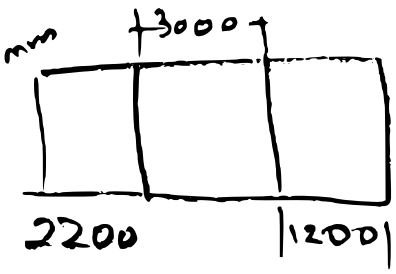
Q5 Home work

@ A cantilever beam span 8m cross sec 250 n 400. if checked for lateral stability beam will fail in. options/ same as question.

(b) isi ka simply supported beam bana lena.

Worksheet Q12. (2) Concept 1507 pg 16.10.11

Ans lo l x + y $\frac{2200}{2}$ b 1000mm



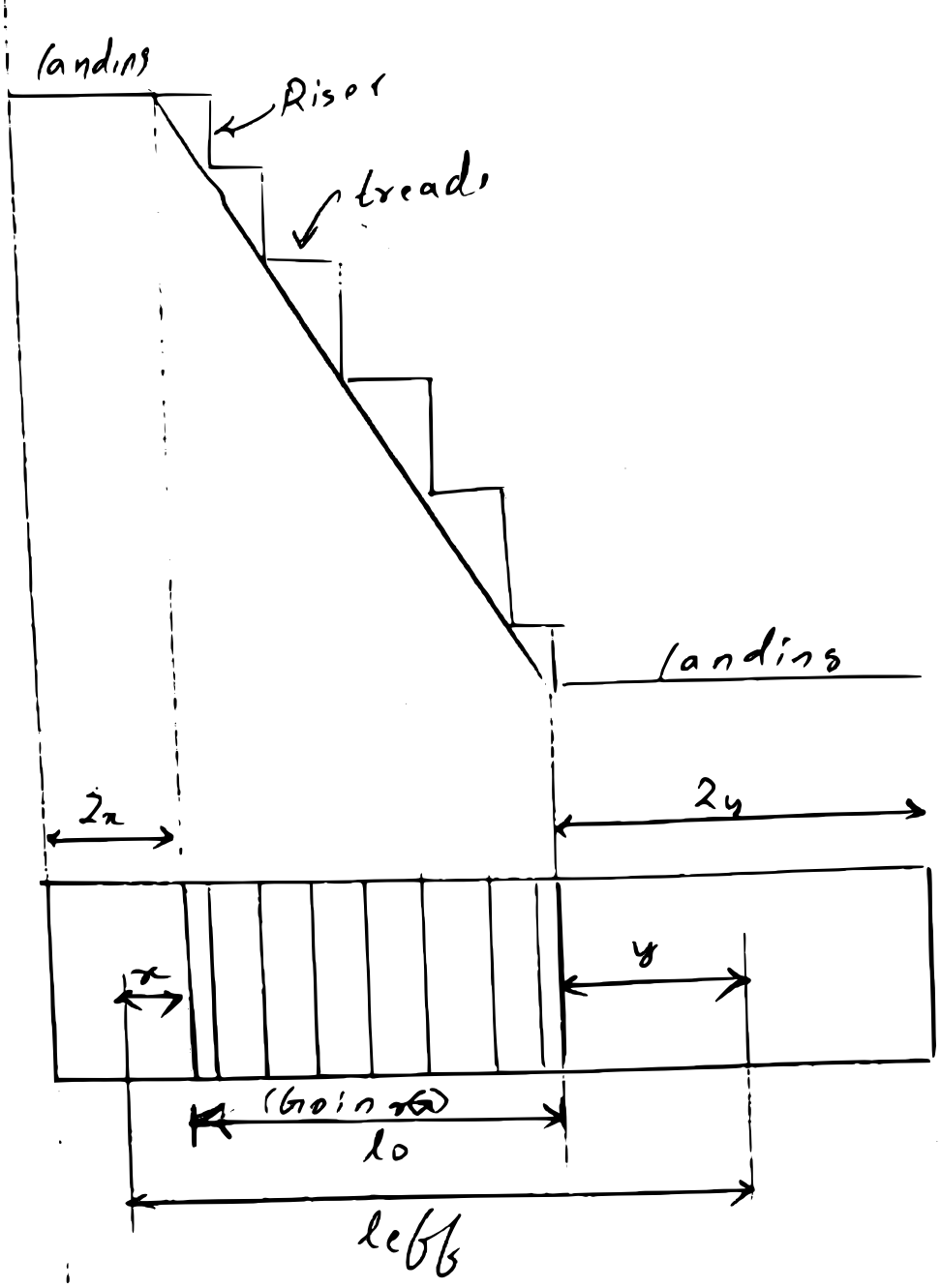
= 3000 + 1000 + 600 ($\frac{1200}{2}$)

= 4600 mm

16.19

Stair Slab or Staircase

side view



Top view

$l_{eff} = l_o + x + y$

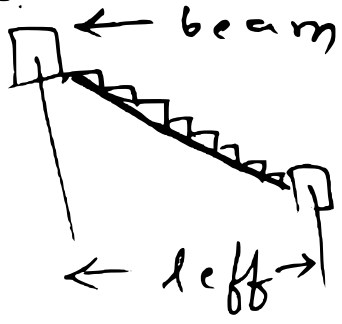
$x \not> 1$

$y \not> 1$

Note stair slab ko running metre ke basis pe ya per metre width pe design krte h isiliye x & y value cannot be more than 1m. if coms more than 1m then consider it as 1 m.

effective span of stair slab case

(a) Beam case



(b) landing beam case

effective span of stair slab

Where stair slabs without stringer beams shall be taken as the following horizontal distances.

(1) When beam is supported at top and bottom on the slab spanning parallel to the raised then $l_{effective}$ = centre to centre distance of beams

(2) When landing slabs are present at top and bottom spanning parallel to raised then $l_{effective} = (l_o + x + y)$

$l_{effective}$ = Going + half the width of landing on each side (cannot be more than 1m)

(3) Where the landing slab span is in the same direction as the stair they shall be considered as acting together to form a single slab and the span is determined as the distance centre to centre of the supporting beam or walls the going being measured horizontally.

17.1

Class 17 — 4 April 2024

Numerical 2 / Question Series

Worksheet no 5 Q 17 (in front of pg 3.9) →

Solution

17.

$b = 250\ mm$

$d = 350\ mm$

$\tau_c = 0.62\ N/mm^2$

$\tau_{c max} = 2.8\ N/mm^2$

$V_u = 200\ kN = 200 \times 10^3\ N$

Vertical stirrups

2 legged 10 mm ϕ Fe 250

Spacing = ?

formula for spacing $\not\vdash 0.75d$, $\not\vdash 300\ mm$, $\not\vdash s_v$

① $0.75\ d = 350 \times 0.75 =$

② 300 mm

③ $s_v = \dfrac{0.87\ f_y\ A_{sv}\ d}{V_{us}}$; $f_y = 250$ ——— (1)

A_{sl}: $2 \times \frac{\pi}{4} \times 10^2 = 157.07\ mm^2$

↙ 2 legged

$V_{us} = V_u - V_c$

$= V_u - \tau_c\, bd$

$V_{us} = 200 \times 10^3 - [0.62 \times 250 \times 350]$

$= 200 \times 10^3 - (54.2 \times 10^3)$

$V_{us} = 145.75 \times 10^3\ N$

Eq^n (1) :-

$S_v = \frac{0.87 \times 250 \times 157.07 \times 350}{145.75 \times 10^3}$

$S_v = 82.04\ mm$

Spacing $\not> 0.75d$: $262.5\ mm$ (350 × 0.75)

300 mm

82.04 mm

Ans: 82.04 mm

$\simeq$ 8.204 cm

$\simeq$ 8.2 cm

$\simeq$ 8.0 cm.

17.3

Worksheet 5 pg 3.9 → Pk pas Qn 0 22

$l_0 = l_{eff} = 6m$

$B_f = ?$

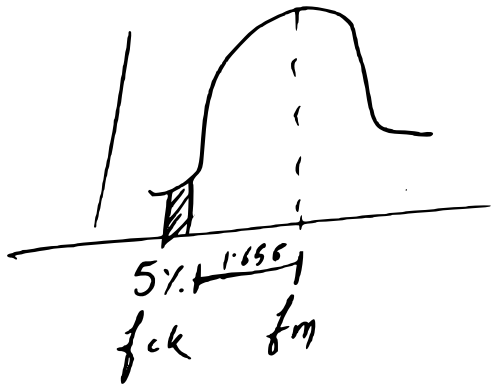
$$B_f = \frac{l_0}{\frac{l_0}{6} + 4} + b_w$$

$$B_f = \frac{6000}{\frac{6000}{1000} + 4} + 300 = 900\ mm$$

Worksheet 5 Qn 23 ka ans

$f_m = f_{ck} + gap$

for 5% failure ~~rate~~ fail wala diagram



f_m : average or mean strength of concrete

50% fall wala digram

$f_m = f_{ck} + gap$ ↗ 70

$\Rightarrow f_m \cdot f_{ck}$ 50% fall

$f_{m\text{-}targ} = 25\ MPa$

Prestressed concrete kya hotah

psc ka code kya hotah?

Team RCM

Poetensioning me kya hotah?

Post tensioning me kya hot h?

Poensioning ka kya method h?

Post tensioniy me kya mthd h?

Min grade of concrete Post tensioniy me kya h?

kyo zaroorri h high strength concpxme

Prestressed concrete (PSC)

IS 1343 : 2012.

Prestress concept uses initial stress application in concrete in order to counterbalance the effect of extern al load partially

Types of prestressing

Pretensioning	Post tensioniv
① Cable is tensione d first & then concreting is done after it.	① Concreting is done first & ducts are left & then cable is tensioned.
② Hoyer's long line method. Example Railway sleepers are manufactured by this method.	② Freessyinet method ③ Lee Mecall system ④ Magnell blaton sys. ⑤ Gifford Vdall system
③ M40 min graded of conc.	③ M30 minimum grade of concrete is used
Necessity of high strength or high grade of concrete in Prestressed concrete.	

17-5

creep loss kamkrne ke liye kya karenge?

modulus of elasticity kaise badhayenge?

high bearing stress ~~kam~~ bear kaise karenge.

kya cable use kr sktc h shrinkg km krne k liye? kaisa cable use kroenge?

kya kya advantage h. Prestress kg RCC pe.

span, load

load, section

section, shear

(a) In order to reduce higher creep loss (of cable) or to increase modulus of elasticity of concrete higher grade of concrete is required.

(b) higher grade of concrete is used to bear high bearing stress which occurs at ends.

(c) Higher grade of steel is also used as cable or wire to reduce creep and shrinkage losses.

Advantages of prestress over RCC.

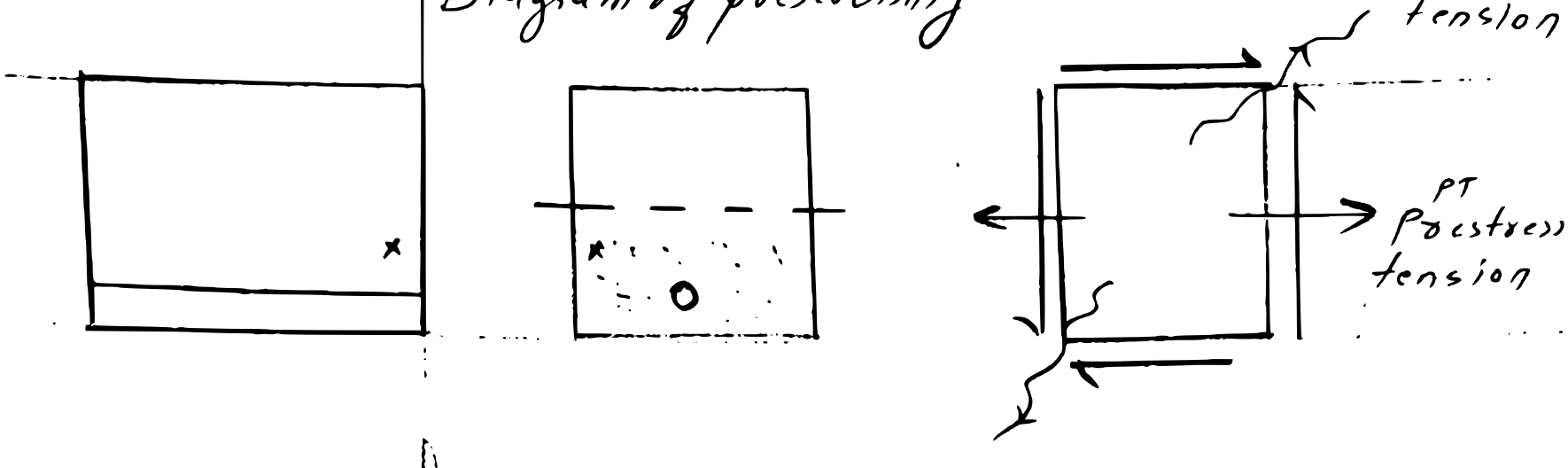
1) Smaller or slender section required for larger span for taking load.

2) In RCC the depth of concrete upto NA participates in taking load whereas in PSC complete depth of section ie complete concrete participates in load carrying.

3) Saving of supporting structure & foundation due to thinner section.

④ Improved shear resistance because principle tension or diagonal tension reduces.

Diagram of prestressing



IS 1343 : Code of practice for prestressed concrete.

What is IS for TaN for?

IS 8112 : Specification for 43 grade OPC

269 : Specification for opc 33 grade

TeN Chess for?

Is 12269 : Specification for 53 grade opc

T.M MeTTa for?

IS 12600 : Specification for low heat portland cement

IS 13311 : Method of Non destructive testing of concrete.

17.7

kya kya assumptions h strain,

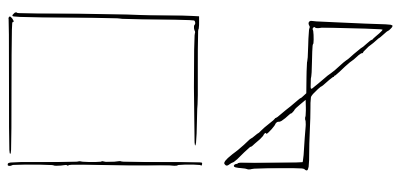
limit state of str me kya kya atah PSC me

limit state of service ability me kya aata h PSC me?

Compression aur max compression me kya farlk hai?

Assumptions

1) Plain section before bending remains plane after bending ie strain/variation is linear along the depth.



linear strain.

2) limit state of ~~collapse~~

collapse/strength/safety

1) flexure Bending Rupture
2) Compression
3) Torsion
4) Shear
5) Tension.

serviceability

1) Deflection
2) Corrosion
3) Maximum compression

naya.

Prestress anal ke kya methods hote h?
SSL

Stress concept me kya kate h?
- Direct stress kya ayega kaisa nature
- Bending stress kya ayega, kaisa nature
→ ps force k karan
→ dead load ke karan
→ live load ke kaaran

Prestress analysis methods

1. Stress concept
2. Strength concept
3. load balancing concept.

Stress concept :

1. Direct stress due to prestressing force

$\sigma_d = \frac{+/load}{+Area}$ [compressive]

2. Bending stress due to prestressing force

prestress me +ve / compression ko lete h

$\sigma_{bp} = \frac{-Pey}{+I}$ [tension at top, compression on bottom]

3. Bending stress due to dead load.

$\sigma_{bd} = \frac{+Mdy}{-I}$ [~~tensio~~ compression @ top, tension @ bottom]

4. Bending stress due to live load

$\sigma_{bl} = \frac{+Mly}{-I}$ [compression @ top, tension @ bottom]

17.9 total stress kya hota h?

5. $\sigma_{total} = \sigma_d + \sigma_{bp} + \sigma_{ld} + \sigma_{ll}$

$$\sigma_{total} = +\left(\frac{P}{A}\right) - \left(\frac{P_e y}{I}\right) + \left(\frac{M_d y}{I}\right) \pm \left(\frac{M_l y}{I}\right)$$

A is crosssection area.

Codal provisions

1) Modulus of elasticity of concrete (IS 1343)

Short term = $5000\sqrt{f_{ck}}$.

long term modulus of elasticity

$E_{c\theta} = \frac{E_c}{1 + \theta\alpha}$ creep coefficient

θ table

Age	θ
7	2.2
28	1.6
1yr	1.1

← table

2) Shrinkage strain

1. Pretensioning

$= 0.0003 = 3 \times 10^{-4}$

Pretensioning me Shrinkage strain kinna hoga

2. Post tensioning

$$= \frac{2 \times 10^{-4}}{\log_{10}(t+2)}$$

t is time in days.

* Note upar me y is extreme fiber distance from NA. (or CG)

I is MOI = $\frac{bd^3}{12}$

Class 18 5 April 2024

σ_{zd} kb zero hoga

σ_{bp} kab zero hoga

σ_{bl} kab zero hoga

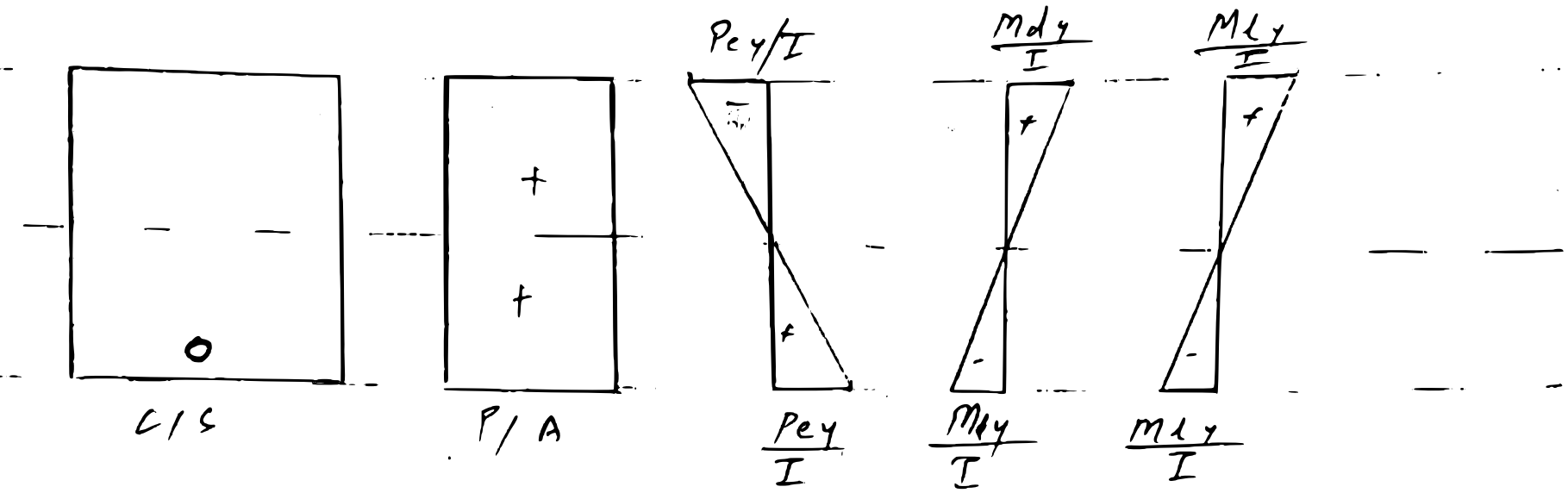
total stress vaun ka fig kaisa hoga?

Cases:

1) Agar question me neglect deadload likha toh σ_{bd} zero ho jayega

2) Agar question mein cable centre me diya ho toh us case me e = 0 hoga hence σ_{bp} bhi zero hoga.

3) Agar question me liveload neglect krne bole toh us case mein $\sigma_{bl} = 0$ ho jayega.



Note: total variation of stress may be of any shape

Ideal condn ke kya parts h?

Initial condn me psc kaisa hoga?

losses kaise honge?

LL Consider hoga kinai?

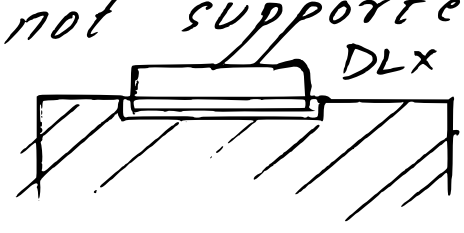
Dead load consider kispe depend karega?

Ideal stress diagram kaisa rehta h?

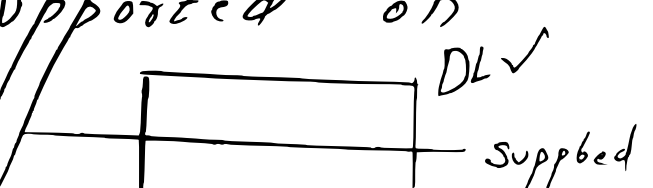
Ideal condition

1) Initial stage or transfer stage.

- full prestressing force is considered
- no losses considered.
- no live load considered.
- Dead load is considered if beam is not supported on supporting bed.



- Dead load is not considered if beam is supported on supporting bed.

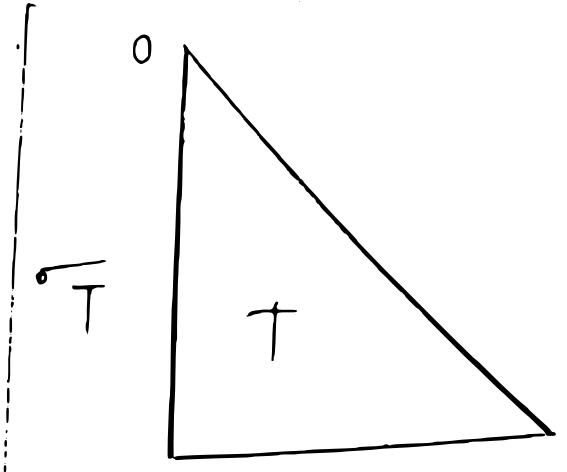


Ideal Stress diagram for initial stage is triangular

0 at top and maximum value at bottom

final stage me kya kya consider karenge.

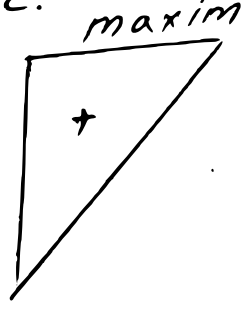
Diagram kaisa hoga?



maximum.

2) At final stage.
- losses are considered
- Deadload is considered
- liveload is considered.

Ideal stress diagram for final stage.



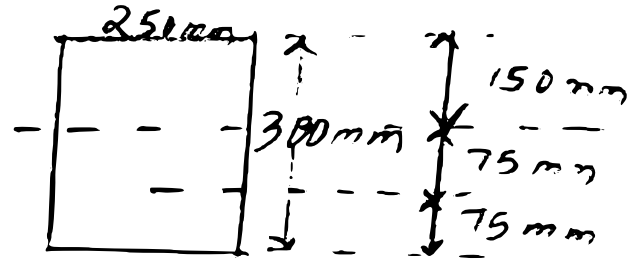
Numerical question SES

Q1 What is the value of total stress at top and bottom fiber in prestress concrete beam if applied force is 500 kN neglect dead load and live load. Cross section is given as 250 X 300 mm and the location of prestressing cable is 75 mm below NA/CG.

18.4 Answer → Given data.

250 x 300 mm

P = 500 kN



Neglecting dead load and live load.

$$\sigma_T = \sigma_d + \sigma_{bp} + \sigma_{yd}^{\circ} + \sigma_{yl}^{\circ}$$

$$\sigma_{Total} = \sigma_{\substack{Top \\ bottom}} = + \frac{P}{A} - \frac{P_e y}{I}$$

$$\sigma_{Top} = + \frac{P}{A} - \frac{P_e y}{I}$$

$$\sigma_{bottom} = + \frac{P}{A} + \frac{P_e y}{I}$$

$$\frac{P}{A} = \frac{500 \times 10^3}{250 \times 300} \frac{N}{mm^2} = 6.67 \ N/mm^2$$

$$\frac{P_e y}{I} = \frac{500 \times 10^3 \times 75 \times \left(\frac{300}{2}\right)}{\left(\frac{250 \times 300^3}{12 \times 6}\right)}$$

$$= \frac{500 \times 10^3 \times 75 \times 6^2}{250 \times 300 \times 300}$$

$$\frac{P_e y}{I} = 10 \ N/mm^2$$

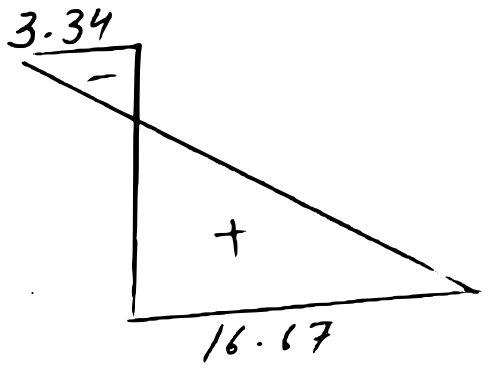
σ_{top} : + 6.67 − 10

: − 3.34 N/mm²

σ_{bottom} = + 6.67 + 10

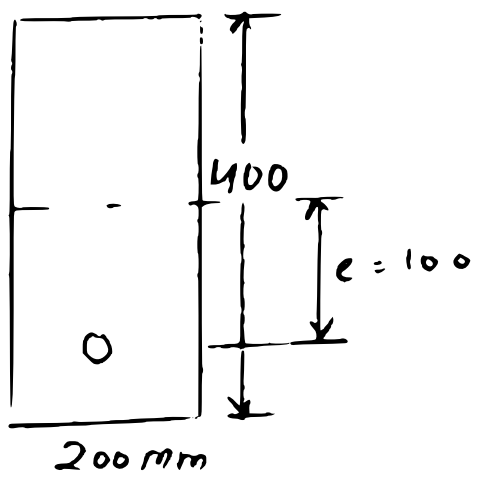
= 16.67 N/mm²

Stress diagram. →



Q2 Given a prestressed rectangular beam of crosssection 200 x 400 mm with prestressing force 400 kN, act at an eccentricity of 100 mm. if the beam is simply supported beam subjected to live load UDL 10 kN/m throughout Beam span 12 m then find out maximum compressive force at support. Neglect dead load.

Ans:



P: 400 kN

neglect dead load

support

max compr stress: ?

Stress ki value support pe nikalne ke liye puchha hai.

For SSB subjected to UDL bending moment at centre is maximum and at support is zero.

ie $M_L = 0 \Rightarrow \sigma_{bL} = 0$.

$$\sigma_{\substack{Top \\ bottom}} : \sigma_d + \sigma_{bp} + \overset{0}{\sigma_{bd}} + \overset{0}{\sigma_{yL}}$$

(neglect dead load) (@ support $M_L = 0$)

$$\sigma_{\substack{Top \\ bottom}} = + \frac{P}{A} - \frac{Pey}{I}$$

Answer after calculation.

$\sigma_{top} : -2.5\ N/mm^2$

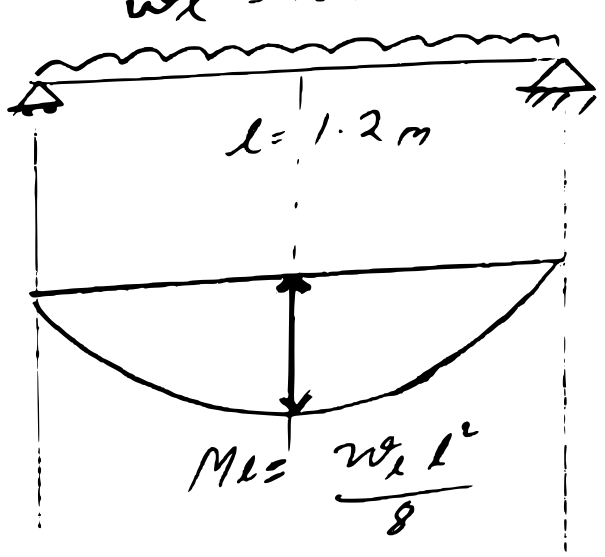
$\sigma_{bottom} : +12.5\ N/mm^2$

Answer

maximum compressive stress : $12.5\ N/mm^2$ @ bottom fiber.

Q2 a → All data of question 2 is same find out max compressive stress (consider mid span of beam)

Answer 2a.



$$M_L = \frac{10 \times 12^2}{8} = \frac{10 \times 12^2}{8} \text{ kN-m}$$

$$= \frac{10 \times 12^2}{8} \times 10^6 \text{ N-mm.}$$

$$\sigma_T = \sigma_d + \sigma_{bp} + \cancel{\sigma_{bd}}^{0} + \sigma_{bl}$$

(neglect) ↓ (mid span)

$$\sigma_d = \frac{P}{A} = \frac{400 \times 10^3}{200 \times 400} = 5 \text{ N/mm}^2$$

$$\sigma_{bp} = \frac{400 \times 10^3 \times 100 \times \frac{400}{2}}{200 \times \frac{400^3}{12}} = 7.5 \text{ N/mm}^2$$

$$\sigma_{bl} = \frac{M_L y}{I} = \frac{10 \times 12^2 \times 10^6 \times \frac{400}{2}}{\frac{200 \times 400^3}{12}}$$

$$= \frac{10 \times 12 \times 12 \times 10^6 \times 6}{200 \times 400 \times 400 \times 8} = \frac{5 \times 27}{4} = 33.7 \text{ N/mm}^2$$

$$\sigma_{top} = +5 - 7.5 + 33.7$$
$$= 31.25 \ N/mm^2$$

$$\sigma_{bottom} = +5 + 7.5 - 33.7$$
$$= -21.25 \ N/mm^2$$

Ans: maximum compressive stress = 31.25 @ top fiber.

IES 2000. Q1a → If all data of question 1 is same if the net losses are 15%. then what is the value of
- ① initial prestresses
- ② final prestresses.

Ans 1a: ① At initial stage no need to consider losses hence answer same as 1 ka hi hoga. $\sigma_{Top} = -3.34 \ N/mm^2$, $\sigma_{bottom} = +16.67 \ N/mm^2$

② At final stage

losses are considered

$P = 500$ kN losses : 15%.

$P = \frac{500}{0.85} \times 10^3$ or $\frac{\text{given load}}{\text{efficiency}} = 588.3$ kN

apan P ko efficiency or gain se divide karenge.

$$\eta = \frac{\text{output}}{\text{input}} \times 100\%$$

15% loss due to efficiency 85% rehi gayi.

Output 500 chahiye tha toh input kitna dena?

$$\eta = \frac{500}{\text{Input}} \times 100\% \quad (\text{Loss} = 15)$$

$$\Rightarrow 85 = \frac{500}{\text{Input}} \times 100\%$$

$$\Rightarrow P = 588.3 \text{ kN}$$

$$\sigma_{Top} = \frac{P}{A} - \frac{Pey}{I} = \frac{588.3 \times 10^3}{250 \times 300} - \frac{588.3 \times 10^3 \times 75 \times \frac{300}{2}}{\frac{250 \times 300^3}{12}}$$

~~$= -5.88 - 11.76$~~

$= -3.92$.

$\sigma_{Bottom} = +19.6 \text{ N/mm}^2$

Ques 2012:
Different prestressing systems in concrete.

(i) By means of hydraulic jack

(ii) By means of thermal methods

(iii) By means of support displacement

(iv) By use of expanding cement.

Ques 2008: Concept. Fully prestressed concrete beams — resist ... part of load by prestress

18.10 IES 2009: Shear resistance of structural concrete member can be improved by

① axial prestressing

② Vertical prestressing

③ Inclined prestressing

IES 2007 → Prestressing force in wire (cable) under thermal stressing can be estimated from - elongation (Δl) of wire and temperature rise (ΔT) $\sigma = \varepsilon E$

$$\sigma = \frac{(\Delta l)}{l} E \quad \left\{ \Delta l = l \alpha \Delta T \right\}$$

$$\sigma = \frac{l \alpha \Delta T}{l} * E = E \alpha (\Delta T)$$

provision 12-1-7-2 Spacing for wires pretensioning

40 mm

ϕ_c

nominal max size of aggregate + 5

Grouped cables: ① groups do se 3 yada te nai ho

② Spacing 40 mm

25 ~~mm~~ Nominal max size of agg + 5

6 April 2024 Class 19th

Loss of (prestress) creep ← due to concrete

Total creep strain : Average of creep strain at beginning and end of the period.

loss of prestress due to shrinkage of concrete :

$\Delta\sigma_{sc} = E_s \cdot \varepsilon_{sc}$

E_s Modulus of elasticity of prestressing steel

$\sigma_{s.c}$ loss of prestress due to shrinkage strain of concrete.

ε_{sc} shrinkage strain of concrete.

ε_{sc} = Pretencioning 3×10^{-4}

Posttencioning $\dfrac{2 \times 10^{-4}}{\log_{10}(t+2)}$

loss of prestress due to relaxation of steel.

This loss may vary with type of steel, initial prestress, age, temperature.

Note: for long term relaxation losses the value of relaxation given in code may be multiplied by 3.

15.2 • Loss of prestress due to elastic shortening of concrete.

1) In case of pretensioning this loss occurs

$\Delta \sigma_c = m \times \sigma_c$

2) In case of post tensioning (ek saath)
- if simultaneous prestressing is done then no loss due to elastic shortening
- if successive prestressing (ek ke baad ek) then - loss due to elastic shortening occurs.

$\frac{1}{2} \sigma_c \times m$.

Note: to prevent earthquake sway damper is used. Example in taipei 101 tuned mass damper is used to protect the building from earthquake. IS 1893 Seismic code.

• Loss of prestress due to slip in Anchorage.
- occurs in post tensioning only

• Loss of prestress due to friction.
- occurs in posttensiong only.

$P_x = P_0 e^{(\mu\alpha + Kx)}$

μ = coeff of friction K = coeff of wave / wobbling

Losses percentage in pretensioning is more than post tensioning due to elastic shortening of concrete.

permissible bearing stress $\leq 0.48 f_{ci} \sqrt{\frac{A_{br}}{A_{pun.}}}$

$\leq 0.8 f_{ck}$

in words... $0.48 \times$ cube strength @ transfer $* \sqrt{\frac{\text{bearing area}}{\text{punching area.}}}$

Side face reinforcement: When depth of web > 500mm, then side face reinforcement is provided.

A_{sf} = 0.1% Area of Web.

A_{sf} = 0.05% of Web Area (On each face)

for Spacing should not be more than 200 mm.

Class 20th 8 April 2024

Worksheet 6 Qn 15

$$P = \left[\begin{matrix}\text{stress in}\\ \text{concrete}\end{matrix} \times \begin{matrix}\text{Area of}\\ \text{concrete}\end{matrix}\right] + \left[\begin{matrix}\text{stress in}\\ \text{steel}\end{matrix} \times \begin{matrix}\text{Area}\\ \text{of steel}\end{matrix}\right]$$

$$P = [\sigma_c A_{concrete}] + [\sigma_s * A_s]$$

Area of concrete : gross area − area of steel.

$$A_{concrete} = (A_c - A_s)$$

Substitute

$$P = \bar{\sigma_c} [A_c - A_s] + \bar{\sigma_s} A_s$$

$$= \bar{\sigma_c} A_c - \sigma_c A_s + \bar{\sigma_s} A_s$$

Modular ratio:

$$m = \frac{E_{steel}}{E_{concrete}} \quad \text{or} \quad m = \frac{E_s}{E_c} = \frac{\bar{\sigma}_{steel}}{\sigma_{concrete}}$$

$$m = \frac{\sigma_s}{\sigma_c} \Rightarrow \bar{\sigma_s} = m\sigma_c$$

$$P = \bar{\sigma_c} A_c - \sigma_c A_c + m\sigma_c A_c$$

$$\Rightarrow P = \bar{\sigma_c} A_c + [m-1]\sigma_c A_s.$$

Note: Iss question me gross area A_c mana hai. Hum concept jab padhle h toh gross area A_g leta hain.

Prestress Loss table

20.2.

Content	Pretensioning	Posttensioning
(1) Elastic Shortening	✓	• simultaneous tensioning ✗ • successive tensioning ✓
(2) Creep of concrete	✓	✓
(3) Shrinkage of concrete	✓	✓
(4) loss due to relaxation of steel	✓	✓
(5) loss due to anchorage slip	✗	✓
(6) loss due to friction	✗	✓
(7) loss due to wobble or wave effect or curvature	✗	✓

Strength concept Prestress

(i) C-line :-

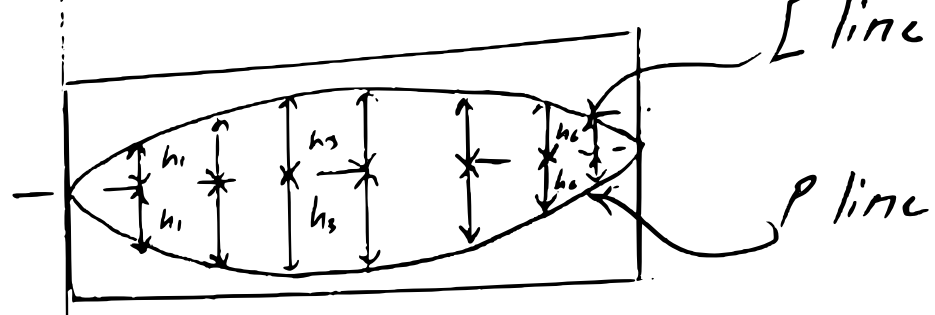
Centre of pressure line or pressure line or thrust line.

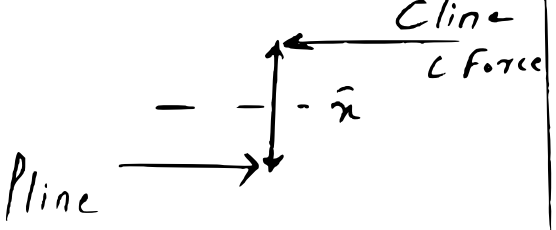
The locus of resultant forces of stress diagram at different cross sections of beam due to all load is called C-line

(ii) P-line :-

Prestressing cable line or cable line.

The profile of cable is called P-line





$M = C\bar{x}$

$= P\bar{x}$

C = C force $\bar{x}$ = shift

P = P force M = moment.

$$\bar{x} = \frac{M}{C} = \frac{M}{P}$$

it is safe between the stress ... can also be termed as given ...

Working Stress Method

1) It is based on stress concept
2) Based on maximum principal stress theory (Rankine)
3) It is deterministic approach.
4) Uses full factor of safety / factor of safety.
5) It does not satisfy serviceability criterion fully.
6) It is based on elastic theory.

permissible, working, modular ratio, these terms are frequently used in WSM. or Aspect ...

Modular ratio:

$$m = \frac{E_{strong}}{E_{weak}} \text{ or } \frac{E_{steel}}{E_{concrete}} \text{ or } \frac{\sigma_{st}}{\sigma_{cbc}} \text{ or } \frac{2\sigma_{st}}{3\sigma_{cbc}}$$

σ_{cbc} = permissible stress in concrete in bending compression. or permissible bend. compr. stress

20.5 ا -

FOS WSM.

Concrete 3

Steel 1.8

Factor of safety for concrete is more than steel because quality control is not proper in case of concrete when compared with Steel.

$\sigma_{cbc} = \frac{f_{ck}}{FOS \text{ of concrete}}$ or $\frac{f_{ck}}{3}$

$\sigma_{st} = f_y / FOS_{steel}$ $= f_y / 1.8$

σ_{st} values (permissile tensile stress in steel)

(i) Fe 250 Mild steel $\sigma_{st} = 140 \text{ N/mm}^2$

(ii) HYSD bar Fe 415 $\sigma_{st} = 230 \text{ N/mm}^2$
Fe 500

σ_{sc} values (permissible compressive stress in steel)

(i) Fe 250 $\sigma_{sc} = 130 \text{ N/mm}^2$

(ii) Fe 415 and above $\sigma_{sc} = 190 \text{ N/mm}^2$

Analysis of Singly reinforced beam by WSM 20.6

Vol 3 RMSE JE, CSE

Assumptions: IPSC, ISRO, Mat Sprcr

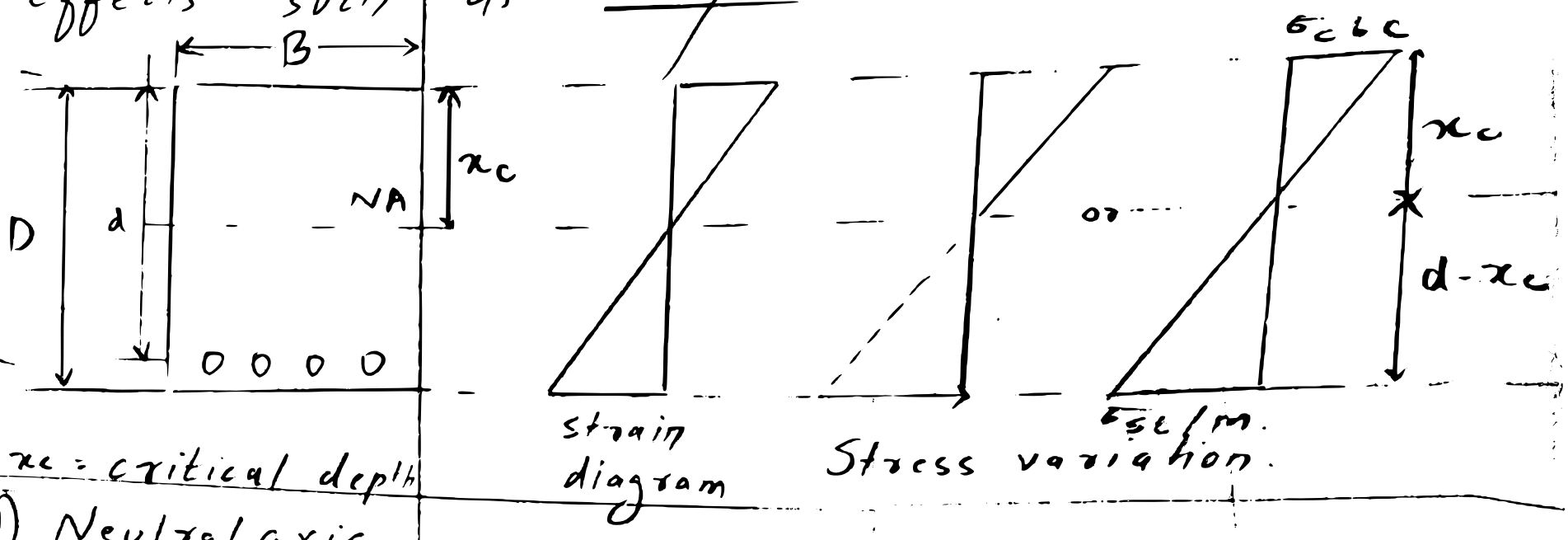
(1) Plane section before bending remains plane after bending. (i.e. strain variation is linear)

(2) All tensile stresses are taken by reinforcement (steel) and none by concrete.

(3) The stress strain relation (stress variation) is assumed straight line (linear) under working load.

(4) Modular ratio is defined $M = \frac{280}{3\sigma_{cbc}}$.

modular ratio M partially takes into account long term effects such as creep.



x_c = critical depth of Neutral axis.

10.7 Determination of critical depth of N/A (x_c)

(i) x_c → using simmilar triangle rule.

$$\frac{\sigma_{cbc}}{x_c} = \frac{\sigma_{st}/m}{(d - x_c)}$$

$$m\sigma_{cbc} \cdot d - m\sigma_{cbc} \cdot x_c = \sigma_{st} \cdot x_c$$

$$m\sigma_{cbc} \times d = \sigma_{st} x_c + m\sigma_{cbc} x_c$$

$$\boxed{\frac{m\sigma_{cbc} \cdot d}{\sigma_{st} + m\sigma_{cbc}} = x_c}$$

$$\left.\begin{matrix}\sigma_{cbc} = c \\ \sigma_{st} = t\end{matrix}\right\} \Rightarrow x_c = \left(\frac{mc}{t + mc}\right) d$$

Note: critical depth of Neutral axis or balanced depth of Neutral axis or limiting depth of neutral axis depends only on grade of steel.

$$x_c = \left[\frac{280/3}{280/3 + \sigma_{st}}\right] d$$

For Fe 250 → $\sigma_{st} = 140\ N/mm^2$

$$\boxed{x_c = 0.4d}$$

For Fe 415 → $\sigma_{st} = 230\ N/mm^2$ ⇒ $\boxed{x_c = .29d}$

Critical depth factor k.

$$k = x_c / d \left(\frac{m\sigma_{cbc}}{m\sigma_{cbc} + \sigma_{st}} \right)$$

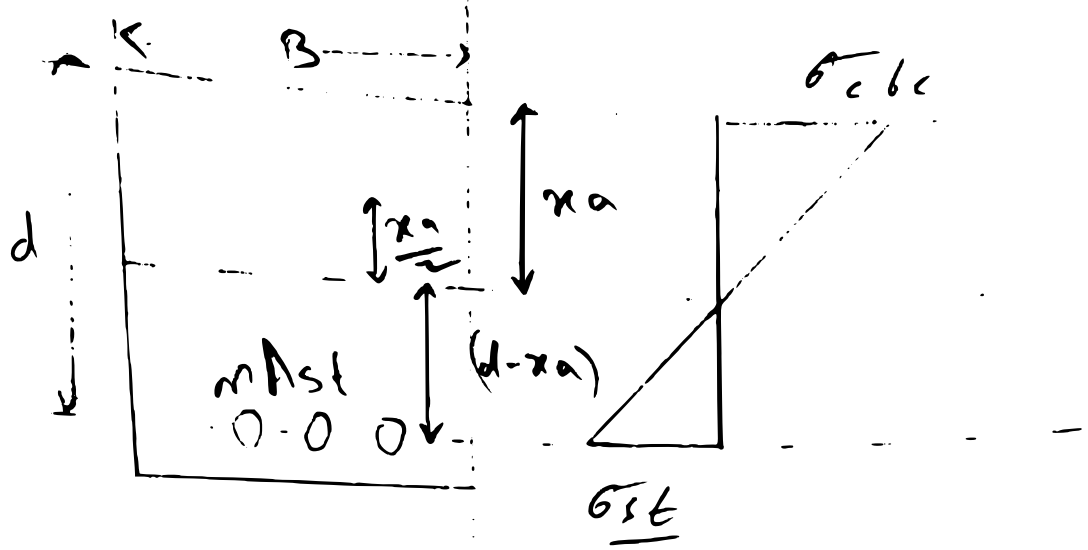
$$k = \left(\frac{mc}{mc + t} \right)$$

Determination of actual depth (x_a) of NA

In WSM actual depth of NA is determined by equating moment of areas about neutral axis. whereas in LSM actual depth of Neutral axis is obtained by equating compressive force = tensile force.

moment of concrete area about NA = moment of steel area about N.A.

21.1 10 April 2024 Class 21



$B x_a \cdot \frac{x_a}{2} = m A_{st} (d - x_a)$

Comparison of sections.

(i) if $x_a < x_c$

URS
under reinforced section.
Moment of resistance is determined from steel/tension side.
Steel reaches its permissible value first and then after it, concrete reaches to its ultimate/permissible value.

(ii) if $x_a = x_c$ MOR is determined from steel or concrete side.
Balanced section

concrete and steel reach permissible limit simultaneously.

(iii) $x_a > x_c$ MOR is determined from concrete side / compression.

Over reinforced section.

Concrete reaches first and then after it steel reaches.

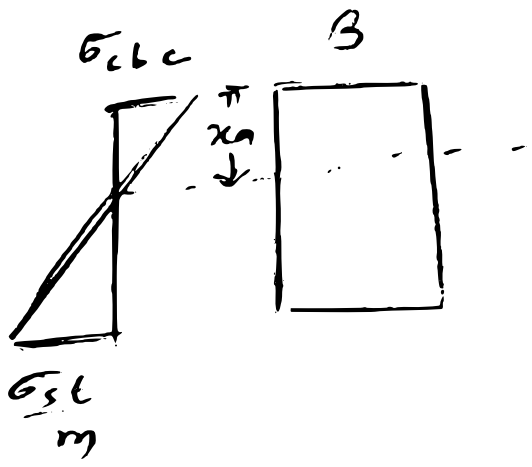
Determination of moment of resistance.

(i) MR = Force x lever arm.

Compression side

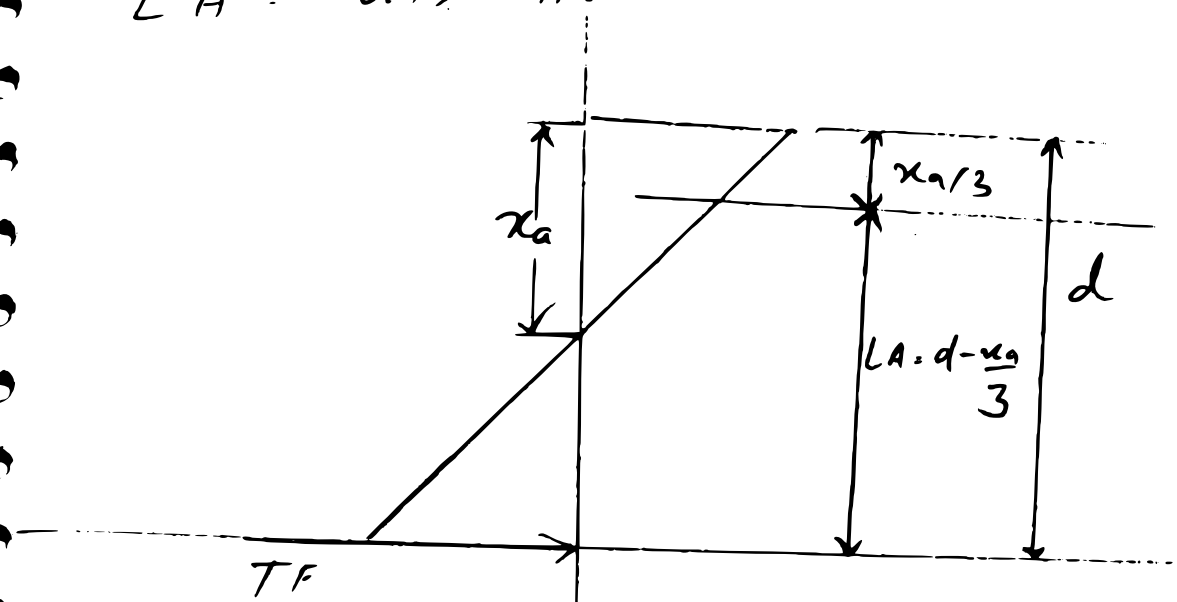
MR = C x LA

MR = stress x area x LA



$MR = \left(\frac{\sigma_{cbc} + 0}{2}\right) * B * x_a * LA$

LA : distance between CF & TF



$LA = (d - 0.33 x_a)$

$MR = \frac{\sigma_{cbc}}{2} B x_a \left(d - \frac{x_a}{3}\right)$

MR = TF × LA : (stress × Area) × LA

(Tension side)

$$MR = \frac{\sigma_{st}}{m} \times m A_{st} \times \left(d - 0 \cdot \frac{x_a}{3}\right)$$

$$MR = (\sigma_{st})\, A_{st} \left[d - \frac{x_a}{3}\right]$$

$$\text{or } M_u = 0.87 f_y A_{st} \left[d - 0.42 x_u\right]$$

Analysis of moment of resistance from compression side.

$$MR = \frac{\sigma_{cbc}}{2} B x_a \left[d - \frac{x_a}{3}\right]$$

for Balanced section.

$x_a = x_c$

$$MR = \frac{\sigma_{cbc}}{2} B x_c \cdot \left[d - \frac{x_c}{3}\right]$$

$$x_c = \left(\frac{m_c}{t + m_c}\right) d$$

$x_c = kd$

$$MR = \frac{\sigma_{cbc}}{2} * B \cdot kd \left[d - \frac{kd}{3}\right]$$

$$MR = \frac{\sigma_{cbc}}{2} * kBd^2 \cdot \left[1 - \frac{k}{3}\right]$$

$\left(1 - \frac{k}{3}\right) = j$ lever arm factor

k : critical depth factor.

$$MR = \frac{\sigma_{cbc}}{2} \cdot Bd^2 \cdot kj$$

$$\Rightarrow MR = \boxed{\frac{\sigma_{cbc} \cdot kj}{2}} Bd^2 = Q \cdot Bd^2$$

$Q = \frac{\sigma_{cbc} \cdot k \cdot j}{2}$

(i) B : width

$$B = \frac{MR}{Qd^2} = \frac{BM}{Qd^2}$$

(ii) $\frac{MR}{QB} = d^2$

$$d = \sqrt{\frac{MR}{QB}}$$

$$= \sqrt{\frac{BM}{QB}}$$

Q. What is the value of moment of resistance for concrete grade 120 & Fe 415

2.3

MR = TF × LA : (stress × Area) × LA

(tension side)

$$MR = \frac{\sigma_{st}}{m} \times m A_{st} \times \left(d - 0.\frac{x_a}{3}\right)$$

$$MR = \sigma_{st} A_{st} \left[d - \frac{x_a}{3}\right]$$

$$^{Lem}M_u = 0.87 f_y A_{st} \left[d - 0.42 x_u\right]$$

Analysis of moment of resistance from compression side.

$$MR = \frac{\sigma_{cbc}}{2} B x_a \left[d - \frac{x_a}{3}\right]$$

for Balanced section.

$$x_a = x_c$$

$$MR = \frac{\sigma_{cbc}}{2} B x_c \cdot \left[d - \frac{x_c}{3}\right]$$

$$x_c = \left(\frac{m_c}{t + m_c}\right) d$$

$$x_c = kd$$

$$M_R = \frac{\sigma_{cbc}}{2} * B \cdot kd \left[d - \frac{kd}{3} \right]$$

$$M_R = \frac{\sigma_{cbc}}{2} * kBd^2 \cdot \left[1 - \frac{k}{3} \right]$$

$\left(1 - \frac{k}{3}\right) = j$ lever arm factor

k : critical depth factor.

$$MR = \frac{\sigma_{cbc}}{2} \cdot Bd^2 \cdot kj$$

$$\Rightarrow MR = \boxed{\frac{\sigma_{cbc} \cdot kj}{2}} Bd^2 = Q \cdot Bd^2$$

$$\therefore Q = \frac{\sigma_{cbc} \cdot k \cdot j}{2}$$

(i) B : width

$$B = \frac{MR}{Qd^2} = \frac{BM}{Qd^2}$$

(ii) $\frac{MR}{QB} = d^2$

$$d = \sqrt{\frac{MR}{QB}}$$

$$= \sqrt{\frac{BM}{QB}}$$

Q. What is the value of moment of resistance for concrete grade 120 & Fe 415

Steel is used. Find out ans in terms of Bd.

M 20
Fe 415

$MR = Q\,Bd^2$

$Q = \frac{\sigma_{cbc}}{2} kj$

$\sigma_{cbc} = \frac{20}{3} \cdot 0.29 \cdot \left(1 - 0.\frac{29}{3}\right)$ | $k = 0.29$ | $j = 1 - \frac{0.29}{3}$

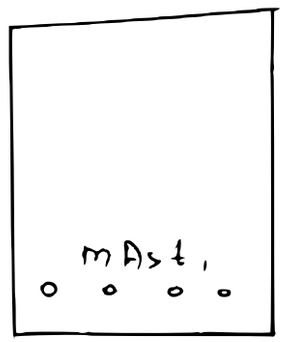
$\Rightarrow Q = 0.87$

$MR = 0.87\,Bd^2$.

Doubly Reinforced Section WSM

When depth and width of beam is restricted or limited in that case compression steel is also provided which is under stressed

When bending moment is more than moment of resistance for limiting or balanced section it will be a doubly rf section.



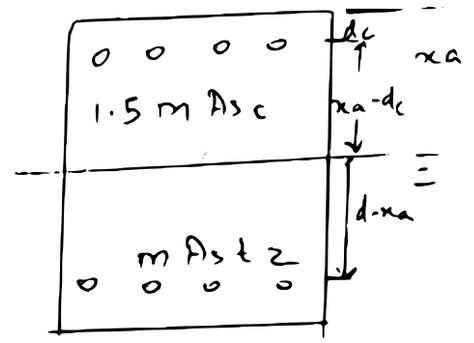
SRS ①

① Moment of compressive area about Neutral Axis MR,

$LM_1 = B x_a \times \frac{x_a}{2}$

• Moment of tensile area about NA.

$TM_1 = mAst\left[d - x_a\right]$

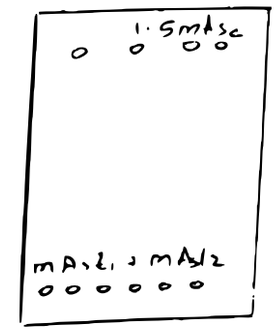


Extra ②

② $CM_2 = [1.5mAsc - Asc]\left(x_a - d_c\right)$

$TM_2 = mAst_2\left[d - x_a\right]$

dc is cover of compression steel.



DRS ③

③ $CM = CM_1 + CM_2$

$CM = B \cdot x_a \cdot \frac{x_a}{2} + (1.5mAsc - Asc)(x_a - dc)$

$TM = TM_1 + TM_2$

$= mAst_1\left[d - x_a\right] + mAst_2\left[d - x_a\right]$

Determination of actual depth of NA x_a for doubly reinforced section [WSM]

$CM = TM$.

2.5 | Steel is used find out ans in terms of Bd.

M 20
Fe 415

$MR = Q\,Bd^2$

$Q = \frac{\sigma_{cbc}}{2} kj$

$\sigma_{cbc} = \frac{20}{3} \cdot 0.29 \cdot \left(1 - 0.\frac{29}{3}\right)$ | $k = 0.29$ | $j = 1 - \frac{0.29}{3}$

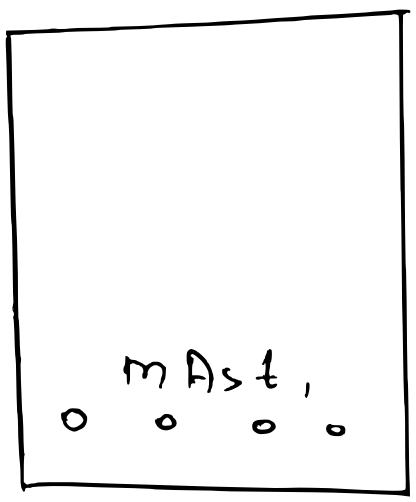
$\Rightarrow Q = 0.87$

$MR = 0.87\,Bd^2$.

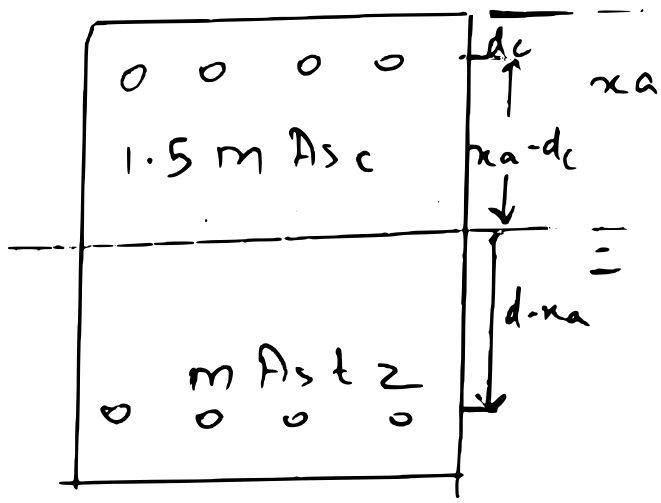
Doubly Reinforced Section WSM

When depth and width of beam is restricted or limited in that case compression steel is also provided which is under stressed

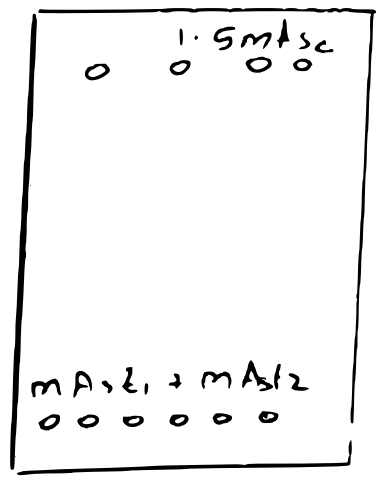
When bending moment is more than moment of resistance for limiting or balanced section it will be a doubly rf section.



SRS
(1)



Extra
(2)



DRS
(3)

(1) moment of compressive area about Neutral Axis MR,

$CM_1 = B x_a * \frac{x_a}{2}$

• Moment of tensile area about NA.

$TM_1 = mAst[d - x_a]$

(2) $CM_2 = [1.5mA_{sc} - A_{sc}](x_a - d_c)$

$TM_2 = mAst_2[d - x_a]$

dc is cover of compression steel.

(3) $CM = CM_1 + CM_2$

$CM = B \cdot x_a \cdot \frac{x_a}{2} + (1.5 mA_{sc} - A_{sc})(x_a - d_c)$

$TM = TM_1 + TM_2$

$= mAst_1[d - x_a] + mAst_2[d - x_a]$

Determination of actual depth of NA x_a for doubly reinforced section. [WSM]

$CM = TM$.

21.7

$$B\frac{x_a}{2}^2 + (1.5m - 1) A_{sc} (x_a - d_c)$$

$$= (m A_{st_1} + m A_{st_2})(d - x_a)$$

$x_a = ?$

Determination of area of steel required for doubly reinforced section.

- $A_{st_1} \longrightarrow MR_{1(T)}$

  $A_{st_2} \longrightarrow MR_{2(T)}$

  $A_{sc} \longrightarrow MR_{2(C)}$

Moment of resistance of first part from Tension side.

$MR_{1(T)} = TF \times LA$

$$= \frac{\sigma_{st}}{m} * m A_{st_1} \left[d - \frac{x_a}{3}\right]$$

$$\boxed{MR_{1(T)} = \sigma_{st} A_{st_1} \left[d - \frac{x_a}{3}\right]}$$

$$MR_{1(T)} = MR_{balanced} = \sigma_{st} A_{st_1} \left[d - \frac{x_a}{3}\right]$$

$$\boxed{A_{st} = MR_{balanced} / \sigma_{st}(d - x_a/3)}$$

$M_{R\,balanced}$ is moment of resistance of balanced section. / limiting section.

$Ast_2 \longrightarrow MR_2\,(T)$

$LA_2 = (d - d_c)$

$MR_{2(T)} = \frac{\sigma_{st}}{\gamma_m} * Ast_2\,(d - d_c)$

$$\boxed{Ast_2 = \frac{MR_{2(T)}}{\sigma_{st}\,(d - d_c)}}$$

Ast_2 : Area of extra tension reinforcement in doubly reinforced section.

$MR_1 : MR_{balanced}$

$MR_1 + MR_2 : MR = BM$

$\downarrow$

$\Rightarrow M_{R\,balanced} + MR_2 : BM$

$MR_2 : BM - MR_{balanced}$

$\downarrow$

$MR_{2(T)}, MR_{2(c)}$

21.1] A_{sc} determination.

$$A_{sc} : M_{R_{2(1)}}$$

$$M_{R_{2(1)}} = \left[1.5\, m A_{sc} - A_{sc}\right] * c' \left[d - d_c\right]$$

$$M_{R_{2(1)}} = \left(1.5m - 1\right) A_{sc} * c' \times \left[d - d_c\right]$$

$$A_{sc} = \frac{M_{R_{2(1)}}}{\left(1.5m - 1\right) c' (d - d_c)} \Rightarrow \boxed{A_{sc} = \frac{BM - MR_{balanced}}{\left(1.5\,m - 1\right)(d - d_c) \times c'}}$$

c' is stress in compression steel.

Shortcut $M : \sigma \times A \times LA$

$\Rightarrow \tilde{A} : \dfrac{M}{\sigma \times LA}$ ← isko use karke easily sab derive kar sakte hain.

Total area of tension steel $\boxed{Ast : Ast_1 + Ast_2}$

WSM LSM comparison.

WSM

① Deterministic

② x_a

$x_{a_{URS}} < x_{a_{BS}} < x_{a_{ORS}}$

③ LA : $d - \frac{x_a}{3}$

$LA_{URS} > LA_{BS} > LA_{ORS}$

LSM.

① Probabalistic

② x_u

$x_{u_{URS}} < x_{u_{BS}} < x_{u_{ORS}}$

ic jaise steel badhega NA depth badhegi aur lever arm ghategा.

③ LA : $d - 0.42 x_u$

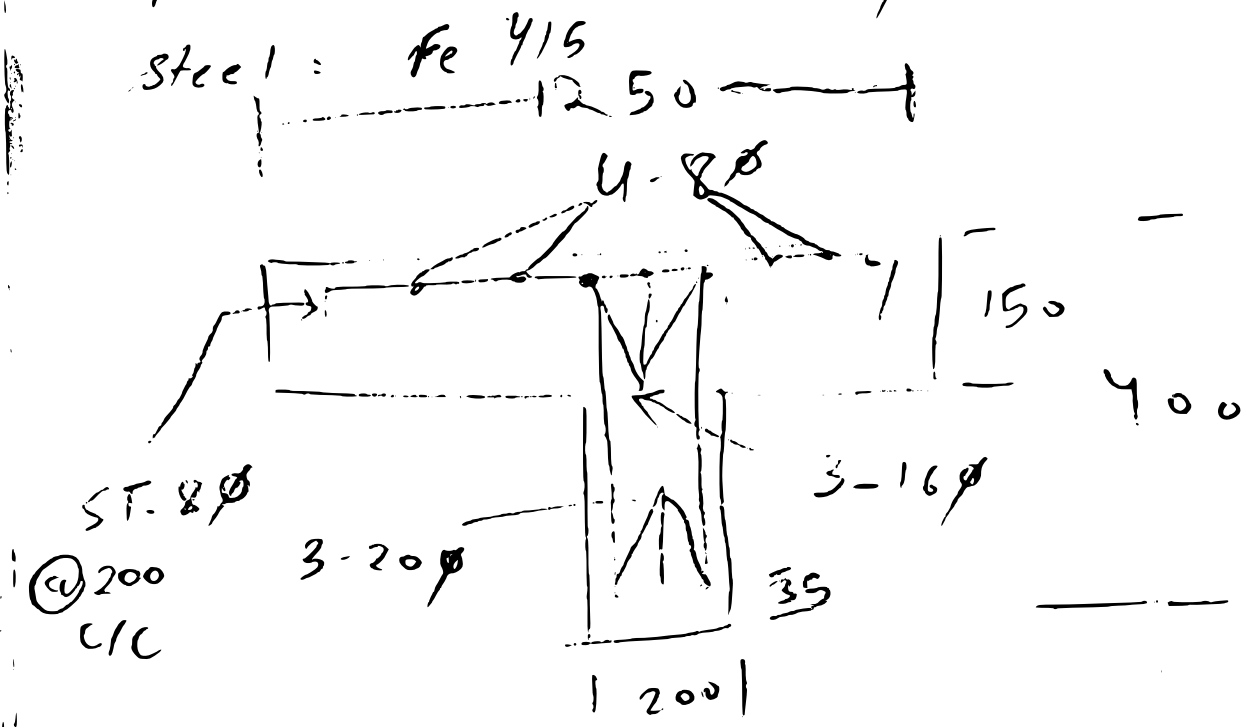
$LA_{URS} > LA_{BS} > LA_{ORS}$

22.1

11 April 2024 class 22

Important Numericals IES

Q. 2.10 for the section span = 6m, conc grade: M20



if the section is checked for serviceability limit state of deflection, it will be found that

→ data insufficient.

Justification: Is jo me limit state of serviceability deflection check karna tha but beam ka support type question me nahi diya hai jisse clear nahi ho pa raha ki beam simply supported hya cantilever hya continuous h isiliye data insufficient answer hoga.

Q.2.11 Max area of compression rf in beam → 0.04 bD.

Q.2.50 In a case of two way slab, the deflection of a slab is → primarily a function of shorter span only.

Q.2.53 A reinforced concrete beam is subjected to following bending moments:

Dead load moment, 20 kN-m

Live load moment 30 kNm

Seismic load moment 10 kNm.

Find The design bending moment for L.S. of collapse is: Limit state of collapse

Ans:

(i) DL moment (DL) : 20 kNm.

(ii) LL/IL moment (IL) : 30 kNm

(iii) SL/EL moment (SL) : 10 kNm

Combinations

(1) DL + IL : 1.5x20 + 1.5x30 : 75 kNm

(2) DL + SL : 1.5x20 + 1.5x10 : 45 kNm

(3) DL + IL + SL : 1.2x20 + 1.2x30 + 1.2x10 : 72 kNm

Ans : 75 kN-m.

22.3 → Q2 53(b) for the same question

What is design bending moment for limit state of serviceability.

Combinations.

DL + IL : 1x20 + 1x30 = 50 kNm

DL + WL/SL : 1x20 + 1x10 = 30 kNm

DL + WL + IL/SL : 1x20 + .8x10 = 52 kNm

1.8x10

Ans: 52 kNm.

Q2.45 The maximum depth of NA for Fe 415 steel grade

→ 0.48d

Q2.46 A simply supported rectangular beam of span 20m is subjected to UDL. The minimum effective depth to check the deflection of beam, when modification factor for tension & compression are 0.9 & 1.10 respectively will be. →

S/s beam

l = 20m

M_t : 0.9

M_c : 1.1

$\frac{l}{d}$: (A) : 20

x span → 10m multiply $\frac{10}{span}$

x Tension of → multiply (m_T)

✓ compression of → multiply (m_c)

$\frac{l}{d} = (A) \times \frac{10}{span} \times m_T \times m_c$

$= \frac{l}{d} = 20 \times \frac{10}{10} \times 0.9 \times 1.1$

$\frac{l^2}{20 \times 10 \times 0.9 \times 1.1} = d \Rightarrow \frac{20^2}{20 \times 10 \times 0.9 \times 1.1} \Rightarrow d = 2.02\,m$

Q.29) How is deflection of RC beams controlled

→ by controlling span to depth ratio.

Q.30. In limit state design moment of resistance of a balanced section using M20 grade concrete Fe 415 is Kbd^2 what is value of K → 2.76.

Q.31 What is maximum spacing between vertical stirrups in RCC beam of rect cross section having effective depth 300 mm

$S_{v\,max} = .75\,d$ adoptable

$\frac{75 \times 300}{300} = 225 \rightarrow$ 225 mm

22.5

Q 2.132 The positive BM coeff @ middle of the end span of a continuous one way slab is —, $\left(\frac{wl}{16} + \frac{wd}{12}\right) l^2$ {option a}

wdl^2	$+1/12$	$-1/10$	$+1/16$	$-1/12$
wll^2	$+1/10$	$-1/9$	$+1/12$	$-1/8$

Q 2.138 se 2.142 ES ka linked question. 2022

A singly reinforced concrete beam with a effective span of 4m has a rectangular cross section with a width of 300mm & depth 550 mm overall. The beam is reinforced with fe 415 & area 250mm^2 @ an effective depth of 500 mm. ~~Consider~~ The S.f wt & with dead load of beam 4 kNm. Consider M15 grade concrete & $\sigma_{cbc} = 5$ Mpa ; $\sigma_{st} = 230$ Mpa

(a) what is BM due to dead load

(b) modular ratio

(c) depth of exit NA

(d) Moment of resistance of section if actual NA is 100 mm

(e) What is permissible live load?

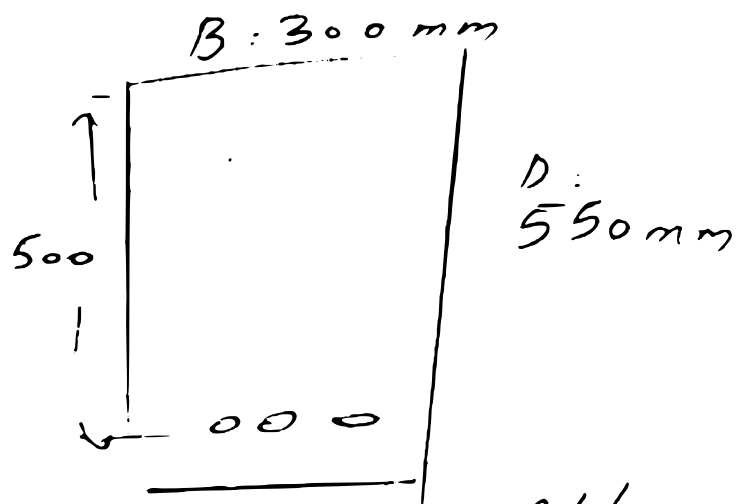
Ans Singly Rf beam

$l_{eff} = 4\,m$

$A_{st} = 250\ mm^2$

Fe 415

m 15.

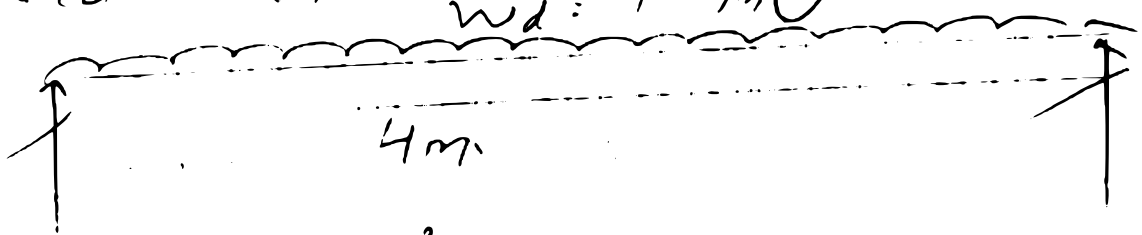


Self weight with dead load: 4 kN/m

$\sigma_{cbc} = 5\ N/mm^2$ $\sigma_{st} = 230\ N/mm^2$

Solution Q 2.138: find bending moment due to dead load.

Ans:



$BM = \frac{wl^2}{8} = \frac{w_d l^2}{8}$

$BM_{dead\ load} = \frac{4 \times 4^2}{8} = 8\ kN\text{-}m.$

$\sim 8000\ N\ m.$ option (2)

Q2.139 What is the modular ratio:

$m = \frac{280}{3\sigma_{cbc}} = \frac{280}{3 \times 5} = \frac{56}{3}$ ≈ Ans. option (c)

Q2.140 What is the depth of crit neut axis.

Fe 415 : $0.29d$ (WSM) $= 0.29 \times 500 = 144.3\ mm.$

Q 2.141 What is the moment of resistance of the section if actual depth of NA is 140 mm →

22.7) $x_a = 100\ mm$

$x_c = 144.33\ mm$

$M_R = ?$

check $x_a < x_c$

$100 < 144.33$

URS

Tension (steel) side:

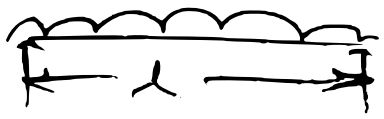
$MR = \sigma_{st} A_{st} \left(d - \frac{x_a}{3}\right) = 230 \times 250 \times \left[500 - \frac{100}{3}\right]$

$= 26.83 \times 10^6\ N\ mm = M_R = 26.83\ kN\ m.$

Q2.142 What is the permissible live load on the beam.

Ans: $M_R = BM_{total\ load} = \frac{w l^2}{8}$

$w^{kN/m} = wd + wl$



$\frac{26.83}{kN\ m} = \frac{w \times 4^2}{8} \Rightarrow w = 13.417\ kN/m.$

$W_{total} = wd + wl \Rightarrow 13.417 = 4 + w_L \Rightarrow wd = 9.42\ \frac{kN}{m}$

Q3.4 2016 if a beam is likely to fail due to high bonding stresses then its bond strength can be increased most economically by using smaller dia bar in correspondingly more numbers

Q3·4+2016 In a rf concrete section (22·8
shear stress distribution is diagramatically
→ Parabolic above NA & rectangular below NA.

Q.3.4c d.
A·347 20m² rectangular beam is of size
230 x 350 mm (eff depth). The factored
shear force acting at a section is
80 KN. If permissible shear stress in
concrete is 0·25 Mpa design shear
stress is
Ans rectangular beam b: 230 mm
d: 350 mm

factored Shear force V_u = 80 KN

Permissible shear stress in concrete is V_c
0.25 Mpa

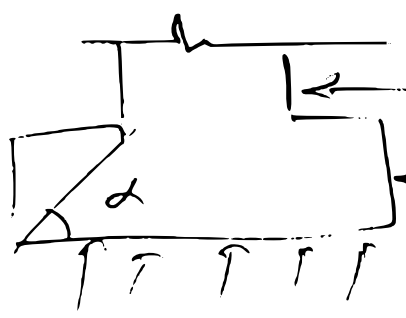
Design shar force = $V_{us} = V_u - V_c$

$= 80 \times 10^3 - \tau_c b d = 80 \times 10^3 - .25 \times 230 \times 350$ N/mm² x mm x mm

$= 59.8 \times 10^3$ N = 59.8 KN. ≃ 60 N. option C

22.9) Design of foundation for a large generator is guided primarily by → frequency

Q. 4.8 A concrete pedestal made of M20 mix shown in fig tand value in this case will be



$q_0 = 3 MPa$

$\tan\alpha \leq 0.9\sqrt{\frac{100 q_0 t}{f_{ck}}}$

$= 0.9 \times \sqrt{\frac{5}{\frac{196 \times 3}{20}} + 1}$

$= 3.6$

$\tan\alpha \leq 3.6$.

Q.4.12 In rc concrete a shear key is provided if → retaining wall is not safe against sliding.

Q.4.13 In an underground warehouse the external walls → are designed as retaining walls with or without surcharge.

Q4.14 lateral ties are provided in rc colums to resit → buckling of longitudnal bars

Q4.15 In an anyxially loaded spirally of short column the concrete inside the core is subjected to → triaxial compression.

04.25 with the additional moments (2 m)
considered for design of slender compression
members in lieu of deflection in x & y dir.

(c) $\frac{Pu_{ex}^2}{2000\,D}$ & $\frac{Pu_{ey}^2}{2000\,b}$

Q. The load carrying capacity of column designed
4.40 [2010]
by WSM is 500kN the ultimate collapse
load of column is → 750 kN

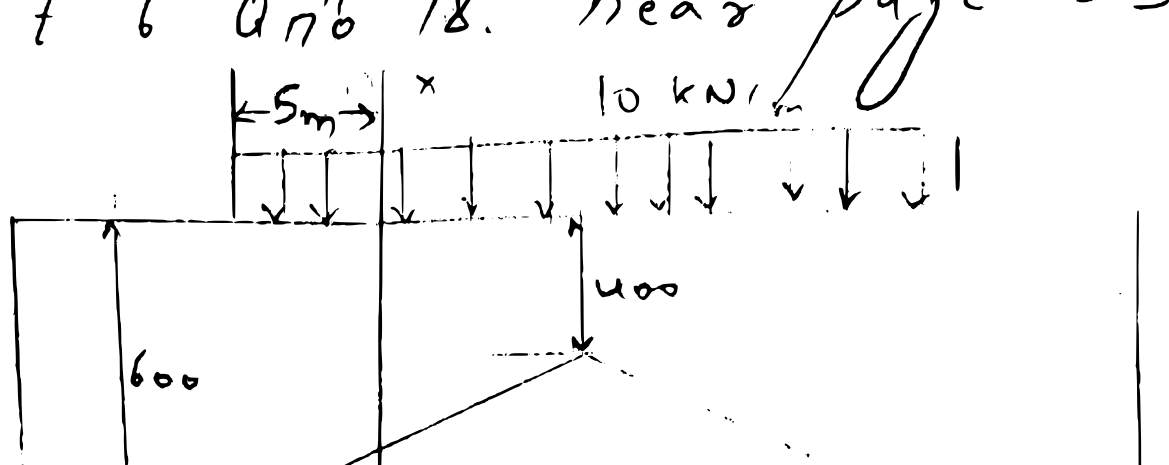
Q. If the load acting on commonly conventional
4.41 [2010]
RC colum increases continuously from zero
to higher magnitudes, the magnitude of
the uniaxial ultimate moment that can be
allowed on the column
→ increases & then decreases.
(Brussler's curve) option (c)
as P increases, M increases then decreases

Q. 436 2009 ans (a)

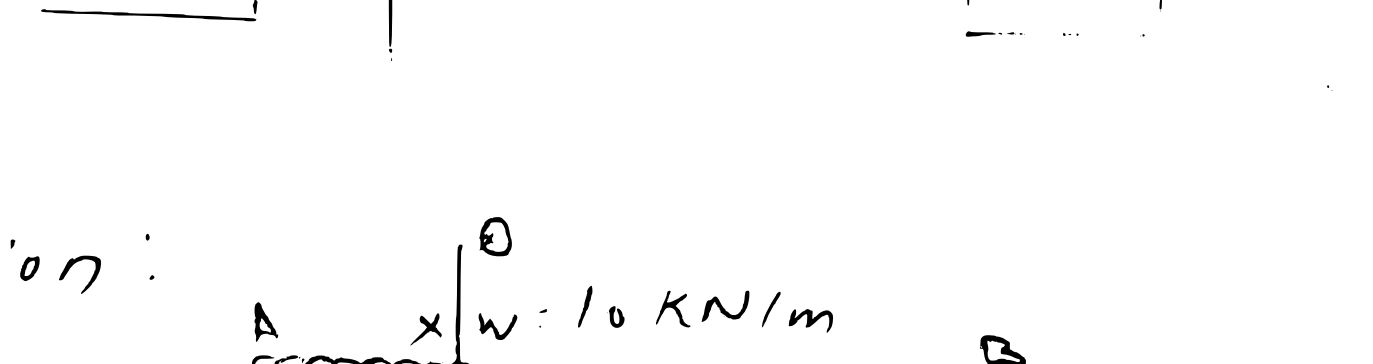
Q. 4.10 Safe load carried
2021. by helically rf colum
→ 1.05 times load carried
by simmilar column with ties

(23-1) 12 December 2024 Class 23

Worksheet 6 Qn⁷o 18. near page 3.9



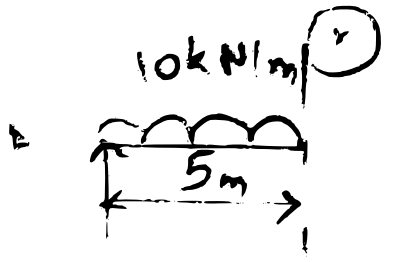
Solution:



Beam me total load = udl x span

= 10 kN/m x 20m

= 200 kN



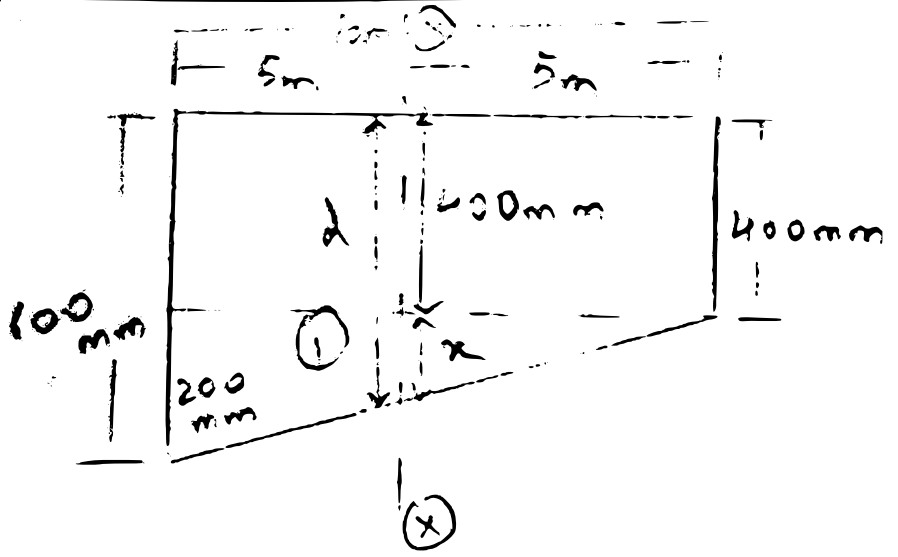
$V_{(x)-(x)} = V_1 = 100 - (10 \times 5)$ kN/m, m

$= 100 - 50 = 50$ kN

$BM_{xx} = M_1 = 100 \times 5 - 10 \times 5 \times 5/2$

kn m

$= 500 - 125 = 375$ kN·m



① Triangle simmilar Δ rule

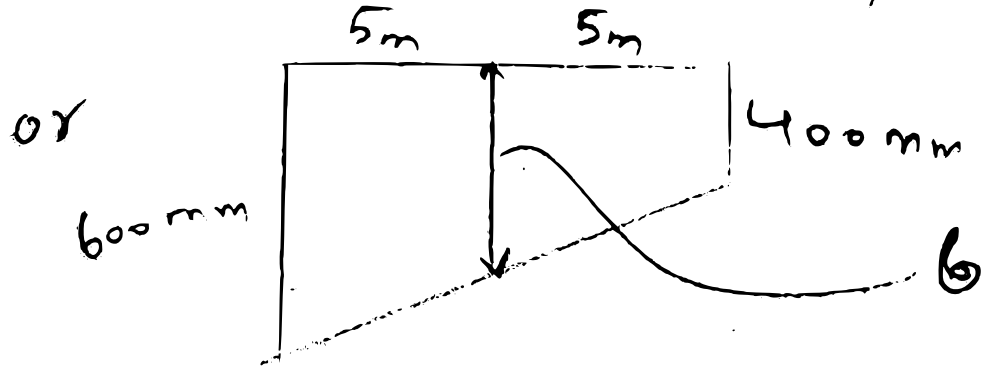
$$\frac{10m}{200mm} = \frac{5m}{x} \Rightarrow x = 100\,mm.$$

$$d = 400 + x$$

$$= 400 + 100$$

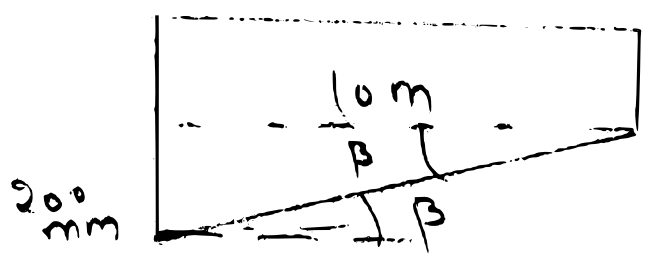
$$= 500\ mm$$

Average concept.



or

$$\frac{600 + 400}{2} = 500\ mm$$



$$\tan\beta = \frac{200mm}{10m}$$

$$= \frac{2\phi\phi}{50\ \cancel{1000}\phi\ mm}$$

$$\Rightarrow \tan\beta = \frac{1}{50}$$

23.3

tapered section.

V_u design $= V_1 \pm \frac{M_1}{d} \tan\beta$.

The given beam is simply supported beam and in simply supported beam. Bending moment decreases towards support and the given beam is tapered beam so depth is increasing towards support.

depth ↑ // BM ↓ ⊕

/

V_u design $= 50\,kN + \frac{375\ kN\cdot m}{0.5\ m} * \frac{1}{50}$

0.5 m
↓
500 mm

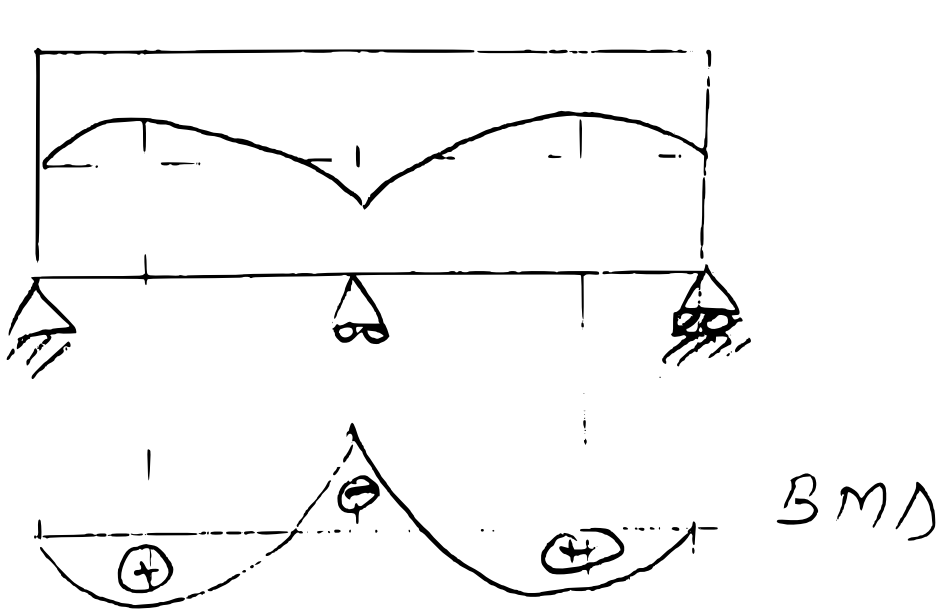
V_u design $= 50 + 15$

$= 65\ kN$

Ve gate Qn tha. Vimp.

Prestress concept : Concordant profile theory :

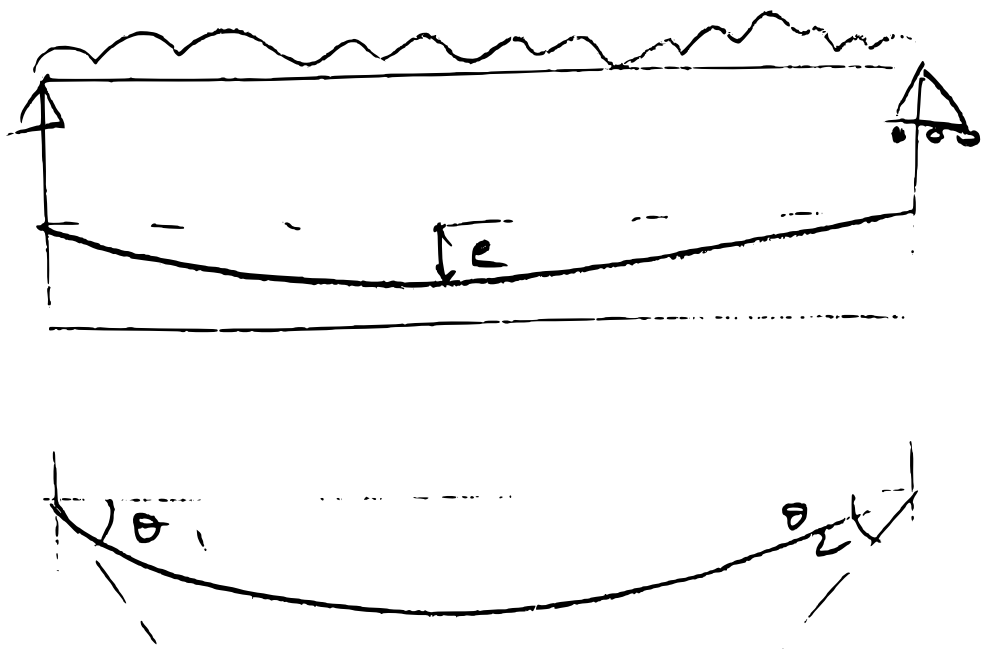
In case of indeterminate beams, in order to keep the maximum stress constant or unchanged the profile of cable should be mirror image of bending moment diagram.



Load Balancing concept :

In this concept cable profile selected in prestressing moment such that transverse component of cable force balances the given type of external load.

23.5



(i) if udl

then slope $\theta = \frac{wl^3}{24EI}$

(ii) if point load

then slope $\theta_1 = \frac{Wl^2}{16EI}$

(iii) Slope value by prestressing force.

$\theta_2 = \frac{Ml}{2EI} = \frac{Pel}{2EI}$

(iv) Net slope $\theta = |\theta_1 - \theta_2|$

net extension or change in length

If prestressing cable $\Delta L = e \cdot \theta$

e is eccentricity

θ is net slope.

Post tensioning

Lee Mccall (Nut bolt) system

Freyssinet → cup cone system